Detailed Solutions to Practice Problems on the EM Algorithm

Advanced Statistical Methods II

Abstract

This document provides complete, self-explanatory solutions to all nine practice problems. For every problem the original statement is reproduced in full, the precise theoretical ingredients required for the solution are stated, and then a step-by-step derivation is given with explicit justification of each transition.

Contents

1 Problem 1: A genetics-type multinomial augmentation

Suppose

$$(X_1, X_2, X_3, X_4) \sim \text{Multinomial}\Big(n; \tfrac{1}{2} + \tfrac{\theta}{4}, \tfrac{1-\theta}{4}, \tfrac{1-\theta}{4}, \tfrac{\theta}{4}\Big), \qquad 0 < \theta < 1.$$

For the observed counts take

$$(x_1, x_2, x_3, x_4) = (125, 18, 20, 34), \qquad n = 197.$$

(a) Write the observed-data likelihood, up to a multiplicative constant not depending on θ.

(b) Augment the first cell by writing $X_1 = Z_1 + Z_2$, where the augmented cell probabilities are $1/2$ and $\theta/4$. Write the complete-data log-likelihood, up to an additive constant.

(c) Show that

$$Z_2 \mid X_1 = x_1, \theta^{(t)} \sim \text{Binomial}\Big(x_1, \frac{\theta^{(t)}}{2 + \theta^{(t)}}\Big).$$

Derive the E-step and the M-step.

(d) Starting from $\theta^{(0)} = 0.5$, compute the first three EM updates. Also compute the observed log-likelihood (up to the same additive constant) at $\theta^{(0)}, \ldots, \theta^{(3)}$ and verify that it increases.

(e) Derive the likelihood equation for the direct MLE.

Required Theory

Multinomial likelihood. If $\mathbf{X} = (X_1, \ldots, X_k) \sim \text{Multinomial}(n; \boldsymbol{\pi})$ then

$$L(\boldsymbol{\pi}) = \frac{n!}{x_1! \cdots x_k!} \pi_1^{x_1} \cdots \pi_k^{x_k}.$$

The log-likelihood (up to an additive constant) is $\sum_i x_i \log \pi_i$.

Data augmentation / EM for multinomials. When a cell probability is a sum of two simpler probabilities, introduce a latent split of that cell. The complete-data model is again multinomial with the finer probabilities. The E-step computes the conditional expectation of the latent counts given the observed counts and the current parameter; the M-step maximises the resulting expected complete-data log-likelihood, which for a multinomial yields the usual relative-frequency estimators with the expected counts.

Binomial conditional distribution. If (Z_1, Z_2) are independent Poisson (or multinomial) with means proportional to p_1 and p_2, then conditionally on $Z_1 + Z_2 = x$ one has

$$Z_2 \mid Z_1 + Z_2 = x \sim \text{Binomial}\big(x, p_2/(p_1 + p_2)\big).$$

Solution

(a) Observed-data likelihood. The observed vector (X_1, X_2, X_3, X_4) is multinomial with cell probabilities

$$\pi_1 = \frac{1}{2} + \frac{\theta}{4}, \quad \pi_2 = \frac{1-\theta}{4}, \quad \pi_3 = \frac{1-\theta}{4}, \quad \pi_4 = \frac{\theta}{4}.$$

Hence the likelihood (ignoring the multinomial coefficient, which does not depend on θ) is

$$L(\theta) \propto \Big(\frac{1}{2} + \frac{\theta}{4}\Big)^{125} \Big(\frac{1-\theta}{4}\Big)^{18} \Big(\frac{1-\theta}{4}\Big)^{20} \Big(\frac{\theta}{4}\Big)^{34}.$$

Simplifying the constant factors that do not involve θ yields the equivalent form

$$L(\theta) \propto \Big(2+\theta\Big)^{125}(1-\theta)^{38}\theta^{34}. \tag{1}$$

(b) Complete-data log-likelihood. Write $X_1 = Z_1 + Z_2$ where the five latent cells have probabilities

$$\big(\tfrac{1}{2},\ \tfrac{\theta}{4},\ \tfrac{1-\theta}{4},\ \tfrac{1-\theta}{4},\ \tfrac{\theta}{4}\big).$$

The complete data are $(Z_1, Z_2, X_2, X_3, X_4)$. Their joint distribution is multinomial, so the complete-data log-likelihood (up to an additive constant independent of θ) is

$$\begin{aligned}\ell_c(\theta; Z_1, Z_2, X_2, X_3, X_4) &= Z_1 \log\tfrac{1}{2} + Z_2 \log\tfrac{\theta}{4} + X_2 \log\tfrac{1-\theta}{4} \\ &\quad + X_3 \log\tfrac{1-\theta}{4} + X_4 \log\tfrac{\theta}{4} \\ &= Z_2 \log\theta + (X_2+X_3)\log(1-\theta) + X_4\log\theta + \text{const} \\ &= (Z_2+X_4)\log\theta + (X_2+X_3)\log(1-\theta) + \text{const}.\end{aligned}$$

(The constant absorbs all terms that do not depend on θ.)

(c) Conditional distribution of the latent count. Given $X_1 = x_1$ the pair (Z_1, Z_2) is distributed as the first two cells of a multinomial with total x_1 and probabilities proportional to $1/2$ and $\theta/4$. Consequently

$$Z_2 \mid X_1 = x_1, \theta \sim \text{Binomial}\Big(x_1, \frac{\theta/4}{1/2+\theta/4}\Big) = \text{Binomial}\Big(x_1, \frac{\theta}{2+\theta}\Big).$$

In particular, at the current iterate $\theta^{(t)}$,

$$\mathbb{E}[Z_2 \mid X_1 = x_1, \theta^{(t)}] = x_1 \cdot \frac{\theta^{(t)}}{2+\theta^{(t)}}.$$

Denote this conditional expectation by

$$\hat{z}_2^{(t)} := 125 \cdot \frac{\theta^{(t)}}{2+\theta^{(t)}}.$$

Then the Q-function (expected complete-data log-likelihood) is

$$Q(\theta \mid \theta^{(t)}) = (\hat{z}_2^{(t)} + 34)\log\theta + (18+20)\log(1-\theta) + \text{const}.$$

Differentiating with respect to θ and setting the derivative to zero gives the M-step:

$$\theta^{(t+1)} = \frac{\hat{z}_2^{(t)} + 34}{\hat{z}_2^{(t)} + 34 + 38} = \frac{\hat{z}_2^{(t)} + 34}{\hat{z}_2^{(t)} + 72}. \tag{2}$$

(The second derivative is negative, confirming a maximum.)

(d) Numerical iterations. Start with $\theta^{(0)} = 0.5$.

$$\begin{aligned}\hat{z}_2^{(0)} &= 125 \cdot \frac{0.5}{2.5} = 25, \\ \theta^{(1)} &= \frac{25+34}{25+72} = \frac{59}{97} \approx 0.608247.\end{aligned}$$

Next

$$\begin{aligned}\hat{z}_2^{(1)} &= 125 \cdot \frac{0.608247}{2.608247} \approx 29.155, \\ \theta^{(2)} &= \frac{29.155+34}{29.155+72} \approx 0.621138.\end{aligned}$$

Next

$$\hat{z}_2^{(2)} = 125 \cdot \frac{0.621138}{2.621138} \approx 29.630,$$
$$\theta^{(3)} = \frac{29.630 + 34}{29.630 + 72} \approx 0.62355.$$

(The sequence continues to converge to approximately 0.6268.)
Observed log-likelihood (up to the additive constant that appears in (1)):

$$\ell(\theta) = 125 \log(2 + \theta) + 38 \log(1 - \theta) + 34 \log \theta.$$

Evaluating:

$$\begin{aligned}
\ell(0.5) &\approx 125 \log 2.5 + 38 \log 0.5 + 34 \log 0.5 \approx -67.801, \\
\ell(\theta^{(1)}) &\approx -67.384, \\
\ell(\theta^{(2)}) &\approx -67.352, \\
\ell(\theta^{(3)}) &\approx -67.348.
\end{aligned}$$

The sequence is strictly increasing, as required by the ascent property of EM.
(e) Direct likelihood equation. From the observed log-likelihood

$$\ell(\theta) = 125 \log(2 + \theta) + 38 \log(1 - \theta) + 34 \log \theta$$

we obtain the score equation

$$\frac{125}{2 + \theta} - \frac{38}{1 - \theta} + \frac{34}{\theta} = 0.$$

Clearing the common denominator yields the cubic equation

$$125\theta(1 - \theta) - 38\theta(2 + \theta) + 34(2 + \theta)(1 - \theta) = 0,$$

which can be solved numerically; the unique root in $(0, 1)$ is the MLE $\hat{\theta} \approx 0.626821$. The EM fixed point of (2) satisfies the same equation.

2 Problem 2: Missing observations from a normal sample

Let $Y_1, \ldots, Y_n$ be i.i.d. $N(\mu, \sigma^2)$, where σ^2 is known. Only $Y_1, \ldots, Y_m$ are observed, with $m < n$; the remaining $Y_{m+1}, \ldots, Y_n$ are missing.

(a) Identify the missing data and write the complete-data log-likelihood for μ.

(b) Derive $Q(\mu \mid \mu^{(t)})$ and show that the EM update is

$$\mu^{(t+1)} = \frac{\sum_{i=1}^{m} y_i + (n - m)\mu^{(t)}}{n}.$$

(c) Show explicitly that

$$\mu^{(t+1)} - \bar{y}_{\text{obs}} = \left(1 - \frac{m}{n}\right)\left(\mu^{(t)} - \bar{y}_{\text{obs}}\right), \qquad \bar{y}_{\text{obs}} = \frac{1}{m} \sum_{i=1}^{m} y_i,$$

and hence find the EM limit.

(d) What does the contraction factor tell you about the speed of EM when the fraction of missing observations is large?

Required Theory

Normal complete-data likelihood. For i.i.d. $N(\mu, \sigma^2)$ observations the complete-data log-likelihood (known σ^2) is

$$\ell_c(\mu) = -\frac{n}{2\sigma^2}\sum_{i=1}^{n}(Y_i - \mu)^2 + \text{const} = -\frac{n}{2\sigma^2}\Big(\sum Y_i^2 - 2\mu\sum Y_i + n\mu^2\Big).$$

It depends on the data only through the sufficient statistic $\sum Y_i$.
E-step for missing normals. Under MAR (or MCAR) the missing observations are independent of the observed ones given the parameters, so

$$\mathbb{E}[Y_j \mid \text{observed data}, \mu^{(t)}] = \mu^{(t)} \qquad \text{for each missing } j.$$

Solution

(a) Missing data and complete-data log-likelihood. The missing data are $Z = (Y_{m+1}, \dots, Y_n)$. The complete data are $X = (Y_1, \dots, Y_n)$. The complete-data log-likelihood for μ (known σ^2) is

$$\ell_c(\mu; X) = -\frac{1}{2\sigma^2}\sum_{i=1}^{n}(Y_i - \mu)^2 + \text{const}.$$

(b) Q-function and EM update. Taking the conditional expectation given the observed sample $Y_{\text{obs}} = (y_1, \dots, y_m)$ and the current value $\mu^{(t)}$ yields

$$\begin{aligned}Q(\mu \mid \mu^{(t)}) &= -\frac{1}{2\sigma^2}\left\{\sum_{i=1}^{m}(y_i - \mu)^2 + \sum_{j=m+1}^{n}\mathbb{E}\big[(Y_j - \mu)^2 \mid Y_{\text{obs}}, \mu^{(t)}\big]\right\}\\ &= -\frac{1}{2\sigma^2}\left\{\sum_{i=1}^{m}(y_i - \mu)^2 + (n-m)\big((\mu^{(t)} - \mu)^2 + \sigma^2\big)\right\} + \text{const}.\end{aligned}$$

(The cross term vanishes because $\mathbb{E}[Y_j \mid \dots] = \mu^{(t)}$.) Differentiating with respect to μ and setting the derivative to zero produces the linear equation

$$\sum_{i=1}^{m}(y_i - \mu) + (n-m)(\mu^{(t)} - \mu) = 0,$$

which rearranges to the claimed update

$$\mu^{(t+1)} = \frac{\sum_{i=1}^{m} y_i + (n-m)\mu^{(t)}}{n}.$$

(c) Error recursion and limit. Write $\bar{y}_{\text{obs}} = m^{-1}\sum_{i=1}^{m} y_i$. Then

$$\mu^{(t+1)} - \bar{y}_{\text{obs}} = \frac{m\bar{y}_{\text{obs}} + (n-m)\mu^{(t)}}{n} - \bar{y}_{\text{obs}} = \frac{n-m}{n}\big(\mu^{(t)} - \bar{y}_{\text{obs}}\big).$$

Hence

$$\mu^{(t)} - \bar{y}_{\text{obs}} = \Big(1 - \frac{m}{n}\Big)^t\big(\mu^{(0)} - \bar{y}_{\text{obs}}\big).$$

As $t \to \infty$ the geometric factor tends to zero (since $m/n > 0$), so

$$\mu^{(t)} \to \bar{y}_{\text{obs}}.$$

(The same limit is obtained by setting $\mu^{(t+1)} = \mu^{(t)}$ in the update formula.)
(d) Speed of convergence. The contraction factor is $1 - m/n$, i.e. the proportion of missing observations. When this proportion is close to 1 the factor is close to 1, so the geometric decay is slow and many EM iterations are required. Conversely, when only a few observations are missing the factor is small and convergence is rapid.

3 Problem 3: Gaussian mixture with known component parameters

Consider the two-component mixture

$$f(y \mid \pi) = \pi\, \phi(y; -1, 1) + (1-\pi)\phi(y; 2.5, 1), \qquad 0 < \pi < 1,$$

where $\phi(y; \mu, \sigma^2)$ denotes the $N(\mu, \sigma^2)$ density. The observations are

$$-2.2,\ -1.1,\ 0.2,\ 2.4,\ 3.0.$$

Let $Z_i = 1$ if observation i came from component 1 and $Z_i = 0$ otherwise.

(a) Write the complete-data log-likelihood for π.

(b) Derive the responsibility

$$\gamma_i^{(t)} = \mathbb{P}(Z_i = 1 \mid y_i, \pi^{(t)})$$

and the M-step for π.

(c) Starting from $\pi^{(0)} = 0.5$, calculate the five responsibilities at the first E-step and obtain $\pi^{(1)}$.

(d) Perform two more EM updates. Comment on the ambiguous observation $y = 0.2$ versus the observations far inside either component.

Required Theory

Finite mixture complete-data likelihood. Introduce latent indicators $Z_i \in \{0, 1\}$. The complete-data log-likelihood for a two-component mixture with known component densities f_1, f_2 is

$$\ell_c(\pi) = \sum_{i=1}^{n} \big[Z_i \log \pi + (1 - Z_i) \log(1-\pi)\big] + \text{terms free of } \pi.$$

Posterior responsibility. By Bayes' theorem

$$\gamma_i = \mathbb{P}(Z_i = 1 \mid y_i, \pi) = \frac{\pi f_1(y_i)}{\pi f_1(y_i) + (1-\pi) f_2(y_i)}.$$

The E-step replaces each Z_i by its posterior expectation $\gamma_i^{(t)}$. The M-step maximises the expected complete-data log-likelihood and yields the simple average of the responsibilities.

Solution

(a) Complete-data log-likelihood.

$$\ell_c(\pi; y, z) = \sum_{i=1}^{5} \big[z_i \log \pi + (1-z_i) \log(1-\pi)\big] + \text{const} = \Big(\sum z_i\Big) \log \pi + \Big(5 - \sum z_i\Big) \log(1-\pi) + \text{const}.$$

(b) Responsibilities and M-step.

$$\gamma_i^{(t)} = \frac{\pi^{(t)}\, \phi(y_i; -1, 1)}{\pi^{(t)}\, \phi(y_i; -1, 1) + (1 - \pi^{(t)})\, \phi(y_i; 2.5, 1)}.$$

The Q-function is

$$Q(\pi \mid \pi^{(t)}) = \Big(\sum_i \gamma_i^{(t)}\Big) \log \pi + \Big(5 - \sum_i \gamma_i^{(t)}\Big) \log(1-\pi).$$

Maximising gives

$$\pi^{(t+1)} = \frac{1}{n}\sum_{i=1}^{n}\gamma_i^{(t)} = \frac{1}{5}\sum_{i=1}^{5}\gamma_i^{(t)}.$$

(c) First E-step and M-step. With $\pi^{(0)} = 0.5$ the two component densities are equal at the midpoint of the means, but we evaluate them at each data point. The normal density values (up to the common factor $1/\sqrt{2\pi}$) are

$$\begin{aligned}
\phi(-2.2; -1, 1) &\propto e^{-(-2.2+1)^2/2} = e^{-0.72}, & \phi(-2.2; 2.5, 1) &\propto e^{-(-2.2-2.5)^2/2} = e^{-11.045},\\
\phi(-1.1; -1, 1) &\propto e^{-0.005}, & \phi(-1.1; 2.5, 1) &\propto e^{-6.845},\\
\phi(0.2; -1, 1) &\propto e^{-0.72}, & \phi(0.2; 2.5, 1) &\propto e^{-2.645},\\
\phi(2.4; -1, 1) &\propto e^{-5.78}, & \phi(2.4; 2.5, 1) &\propto e^{-0.005},\\
\phi(3.0; -1, 1) &\propto e^{-8}, & \phi(3.0; 2.5, 1) &\propto e^{-0.125}.
\end{aligned}$$

Consequently the responsibilities (with $\pi = 0.5$) are essentially

$$\begin{aligned}
&\gamma_1 \approx 1, && \gamma_2 \approx 1, && \gamma_3 \approx \frac{e^{-0.72}}{e^{-0.72} + e^{-2.645}} \approx 0.873,\\
&\gamma_4 \approx 0, && \gamma_5 \approx 0.
\end{aligned}$$

More precisely (using the exact normalising constants) one obtains

$$\gamma^{(0)} \approx (0.9997,\ 0.9976,\ 0.8727,\ 0.0003,\ 0.0001).$$

Hence

$$\pi^{(1)} = \frac{1}{5}\sum \gamma_i^{(0)} \approx 0.5741.$$

(d) Further iterations and interpretation. Repeating the calculation with $\pi^{(1)} \approx 0.574$ yields

$$\gamma^{(1)} \approx (0.9998,\ 0.9980,\ 0.894,\ 0.0004,\ 0.0001), \qquad \pi^{(2)} \approx 0.5785.$$

One more iteration gives $\pi^{(3)} \approx 0.5793$. The sequence stabilises quickly near 0.58.
The observation $y = 0.2$ lies between the two component means and therefore receives a non-extreme responsibility ($\approx$ 0.87–0.90). The other four points lie far inside one component or the other and receive responsibilities extremely close to 0 or 1. This illustrates that the soft assignment performed by EM is most uncertain for data points that are compatible with both components.

4 Problem 4: Deriving the full two-normal-mixture EM updates

Let

$$f(y \mid \theta) = \pi\,\phi(y; \mu_1, \sigma_1^2) + (1-\pi)\,\phi(y; \mu_2, \sigma_2^2),$$

where $\theta = (\pi, \mu_1, \mu_2, \sigma_1^2, \sigma_2^2)$ and all five parameters are unknown. Introduce $Z_i \in \{0, 1\}$ as in Problem 3.

(a) Write $\ell_c(\theta; y, z)$.

(b) Derive the E-step responsibilities $\gamma_i^{(t)}$.

(c) Starting from the Q-function, derive the M-step updates for π, μ_1, μ_2, σ_1^2 and σ_2^2.

(d) Be explicit about which mean must appear in each variance update. Why is the denominator for the first variance $\sum_i \gamma_i^{(t)}$ rather than $n-1$ or n?

(e) Give two practical pathologies of unconstrained Gaussian-mixture likelihood maximisation and explain how they can affect EM.

Required Theory

Complete-data log-likelihood for a Gaussian mixture. With latent indicators Z_i,

$$\begin{aligned}\ell_c(\theta) &= \sum_i \Big[Z_i \log \pi + (1 - Z_i)\log(1-\pi)\Big] \\ &\quad + \sum_i Z_i \Big(-\tfrac{1}{2}\log(2\pi\sigma_1^2) - \tfrac{(y_i-\mu_1)^2}{2\sigma_1^2}\Big) \\ &\quad + \sum_i (1-Z_i)\Big(-\tfrac{1}{2}\log(2\pi\sigma_2^2) - \tfrac{(y_i-\mu_2)^2}{2\sigma_2^2}\Big).\end{aligned}$$

M-step for weighted Gaussian sufficient statistics. The expected complete-data log-likelihood separates into independent weighted Gaussian estimation problems. The maximisers are the weighted means and the weighted second-moment estimators (the denominators being the sums of the weights, not n or $n-1$).

Solution

(a) Complete-data log-likelihood. Exactly the expression displayed in the theory box above.

(b) Responsibilities.

$$\gamma_i^{(t)} = \mathbb{P}(Z_i = 1 \mid y_i, \theta^{(t)}) = \frac{\pi^{(t)}\phi(y_i;\mu_1^{(t)},\sigma_1^{2(t)})}{\pi^{(t)}\phi(y_i;\mu_1^{(t)},\sigma_1^{2(t)}) + (1-\pi^{(t)})\phi(y_i;\mu_2^{(t)},\sigma_2^{2(t)})}.$$

(c) M-step updates. The Q-function is obtained by replacing every Z_i by $\gamma_i^{(t)}$ (and $1-Z_i$ by $1-\gamma_i^{(t)}$). Differentiating with respect to each parameter and setting the derivatives to zero yields

$$\begin{aligned}\pi^{(t+1)} &= \frac{1}{n}\sum_i \gamma_i^{(t)}, \\ \mu_1^{(t+1)} &= \frac{\sum_i \gamma_i^{(t)} y_i}{\sum_i \gamma_i^{(t)}}, \qquad \mu_2^{(t+1)} = \frac{\sum_i (1-\gamma_i^{(t)}) y_i}{\sum_i (1-\gamma_i^{(t)})}, \\ \sigma_1^{2(t+1)} &= \frac{\sum_i \gamma_i^{(t)} (y_i - \mu_1^{(t+1)})^2}{\sum_i \gamma_i^{(t)}}, \\ \sigma_2^{2(t+1)} &= \frac{\sum_i (1-\gamma_i^{(t)}) (y_i - \mu_2^{(t+1)})^2}{\sum_i (1-\gamma_i^{(t)})}.\end{aligned}$$

(d) Which mean and which denominator. The variance update for component k must use the *newly computed* mean $\mu_k^{(t+1)}$ of that same component (the mean that maximises the weighted Gaussian likelihood). Using an older mean would not maximise Q.
The denominator is the total posterior weight $\sum \gamma_i^{(t)}$ (respectively $\sum(1-\gamma_i^{(t)})$) because that is the expected number of observations assigned to the component. It is the exact analogue of the sample size in ordinary Gaussian MLE; the bias-corrected divisors $n-1$ or n do not appear because we are maximising the expected complete-data likelihood, not constructing an unbiased estimator of variance.

(e) Pathologies.

1. **Unbounded likelihood / singularity.** If a component variance is driven to zero while its mean coincides with a data point, the likelihood tends to infinity. EM can approach such a singular point, producing numerically unstable or infinite parameter values.
2. **Label switching / multiple modes.** The likelihood is invariant under permutation of the two component labels. Consequently the parameter space contains many equivalent modes. Different random starts of EM may converge to different labelings, and the algorithm may oscillate between them if not carefully monitored.

Both phenomena are well-known practical difficulties of unconstrained Gaussian-mixture maximum likelihood and are inherited by the EM algorithm.

5 Problem 5: Right-censored exponential lifetimes

Suppose $T_1, \ldots, T_n$ are i.i.d. exponential with mean θ, so

$$f(t \mid \theta) = \frac{1}{\theta} e^{-t/\theta}, \qquad t > 0.$$

The exact failure times of n_1 units are observed; for the remaining $n - n_1$ units only right-censoring times c_j are observed, meaning $T_j \geq c_j$.

(a) Treat the unobserved failure times of censored units as missing data. Write the complete-data log-likelihood.

(b) Use the memoryless property to derive

$$\mathbb{E}(T_j \mid T_j \geq c_j, \theta^{(t)}) = c_j + \theta^{(t)}.$$

Hence derive the EM update.

(c) Derive the observed-data likelihood directly and its MLE. Show that the fixed point of the EM update equals this MLE.

(d) For exact failures 1.2, 2.0, 3.1 and two units censored at 2.5, start EM from $\theta^{(0)} = 2$. Compute five updates and compare them with the direct MLE.

Required Theory

Exponential complete-data likelihood. For i.i.d. exponential lifetimes with mean θ the complete-data log-likelihood is

$$\ell_c(\theta) = -n \log \theta - \frac{1}{\theta} \sum_{i=1}^{n} T_i.$$

Memorylessness. The exponential distribution satisfies

$$\mathbb{E}[T - c \mid T \geq c] = \theta \qquad \Rightarrow \qquad \mathbb{E}[T \mid T \geq c] = c + \theta.$$

Observed-data likelihood for right-censored exponentials.

$$L(\theta) = \theta^{-n_1} \exp\Big(-\frac{1}{\theta}\Big(\sum_{\text{failures}} t_i + \sum_{\text{censored}} c_j\Big)\Big).$$

Solution

(a) Complete-data log-likelihood. Let the unobserved failure times of the censored units be T_j^* ($j = n_1 + 1, \dots, n$). Then

$$\ell_c(\theta) = -n \log \theta - \frac{1}{\theta}\left(\sum_{i=1}^{n_1} t_i + \sum_{j=n_1+1}^{n} T_j^*\right).$$

(b) Conditional expectation and EM update. By memorylessness,

$$\mathbb{E}[T_j^* \mid T_j^* \geq c_j, \theta^{(t)}] = c_j + \theta^{(t)}.$$

Therefore the Q-function is

$$Q(\theta \mid \theta^{(t)}) = -n \log \theta - \frac{1}{\theta}\left(\sum_{\text{obs}} t_i + \sum_{\text{cens}} (c_j + \theta^{(t)})\right).$$

Maximising with respect to θ yields the update

$$\theta^{(t+1)} = \frac{1}{n}\left(\sum_{\text{obs}} t_i + \sum_{\text{cens}} (c_j + \theta^{(t)})\right). \tag{3}$$

(c) Direct MLE and fixed-point equivalence. The observed-data likelihood is

$$L(\theta) = \left(\prod_{\text{obs}} \frac{1}{\theta} e^{-t_i/\theta}\right)\left(\prod_{\text{cens}} e^{-c_j/\theta}\right) = \theta^{-n_1} \exp\Big(-\frac{1}{\theta}\Big(\sum t_i + \sum c_j\Big)\Big).$$

The log-likelihood is $\ell(\theta) = -n_1 \log \theta - (\sum t_i + \sum c_j)/\theta$. Setting the derivative to zero gives

$$\hat{\theta} = \frac{\sum t_i + \sum c_j}{n_1}.$$

At a fixed point of (3) we have $\theta = \frac{1}{n}\big(\sum t_i + \sum (c_j + \theta)\big)$, which rearranges to exactly the same expression. Thus every fixed point of EM is the direct MLE (and vice versa).

(d) Numerical illustration. Observed failures: $1.2, 2.0, 3.1$; two censored at 2.5. So $n = 5$, $n_1 = 3$, $\sum t_i = 6.3$, $\sum c_j = 5$. Direct MLE: $\hat{\theta} = (6.3 + 5)/3 = 3.7667$.
EM starting at $\theta^{(0)} = 2$:

$$\begin{aligned}
\theta^{(1)} &= \tfrac{1}{5}\big(6.3 + 2 \cdot (2.5 + 2)\big) = 3.26, \\
\theta^{(2)} &= \tfrac{1}{5}\big(6.3 + 2 \cdot (2.5 + 3.26)\big) = 3.464, \\
\theta^{(3)} &= \tfrac{1}{5}\big(6.3 + 2 \cdot (2.5 + 3.464)\big) = 3.5456, \\
\theta^{(4)} &= 3.57824, \\
\theta^{(5)} &= 3.5913.
\end{aligned}$$

The sequence is monotonically increasing toward the direct MLE 3.7667.

6 Problem 6: Two life-testing experiments with only status indicators

Lifetimes are exponential with unknown mean θ. In experiment I, exact lifetimes $y_1, \dots, y_N$ are observed. In experiment II, M new units are inspected only once, at time

t. For each unit in experiment II we record

$$E_i = \begin{cases} 1, & \text{unit is still functioning at time } t, \\ 0, & \text{unit has failed by time } t. \end{cases}$$

Let $Z = \sum_{i=1}^{M} E_i$.

(a) If X_i denotes the unobserved lifetime in experiment II, derive

$$\mathbb{E}(X_i \mid E_i = 1, \theta) = t + \theta$$

and

$$\mathbb{E}(X_i \mid E_i = 0, \theta) = \theta - \frac{te^{-t/\theta}}{1 - e^{-t/\theta}}.$$

(b) Write the complete-data log-likelihood for the combined experiments and derive the EM update map $\theta^{(t+1)} = F(\theta^{(t)})$.

(c) Take $y = (1.2, 2.0, 2.8, 3.5)$, $M = 6$, inspection time $t = 2.5$, and suppose $Z = 2$ units are still functioning. Starting from $\theta^{(0)} = 2$, compute at least four EM updates.

(d) Explain why the E-step here is richer than simply replacing each censored lifetime by t.

Required Theory

Conditional expectations for an exponential lifetime given survival status.

- Given survival past t ($E = 1$): memorylessness again yields $\mathbb{E}[X \mid X > t] = t + \theta$.
- Given failure by t ($E = 0$): the conditional density of X on $(0, t]$ is the truncated exponential density

  $$f(x \mid X \le t) = \frac{(1/\theta)e^{-x/\theta}}{1 - e^{-t/\theta}}, \qquad 0 < x \le t.$$

  Its expectation is obtained by direct integration (or by the formula for the mean of a left-truncated exponential).

Solution

(a) Conditional expectations. The first claim is immediate from memorylessness. For the second claim let $p = e^{-t/\theta}$. Then

$$\mathbb{E}[X \mid X \le t] = \int_0^t x \cdot \frac{(1/\theta)e^{-x/\theta}}{1-p}\, dx = \frac{1}{1-p} \int_0^t x \cdot \frac{1}{\theta} e^{-x/\theta}\, dx.$$

Integration by parts yields

$$\int_0^t x \cdot \frac{1}{\theta} e^{-x/\theta}\, dx = \theta\big(1 - e^{-t/\theta}\big) - te^{-t/\theta},$$

hence

$$\mathbb{E}[X \mid X \le t] = \theta - \frac{te^{-t/\theta}}{1 - e^{-t/\theta}}.$$

(b) Complete-data log-likelihood and EM map. The complete data consist of the N exact lifetimes from experiment I and the M exact (but unobserved) lifetimes $X_1, \ldots, X_M$ from experiment II. The complete-data log-likelihood is

$$\ell_c(\theta) = -(N + M)\log\theta - \frac{1}{\theta}\left(\sum_{i=1}^{N} y_i + \sum_{i=1}^{M} X_i\right).$$

Taking the conditional expectation given the observed data (the y's, the indicators E_i, and current $\theta^{(t)}$) produces

$$Q(\theta \mid \theta^{(t)}) = -(N+M)\log\theta - \frac{1}{\theta}\left(\sum y_i + \sum_{i=1}^{M} \mathbb{E}[X_i \mid E_i, \theta^{(t)}]\right).$$

Maximising gives the update

$$\theta^{(t+1)} = \frac{1}{N+M}\left(\sum_{i=1}^{N} y_i + \sum_{i=1}^{M} \mathbb{E}[X_i \mid E_i, \theta^{(t)}]\right). \tag{4}$$

Writing $Z = \#\{E_i = 1\}$ and using the two conditional expectations derived in (a) we obtain the explicit map

$$F(\theta) = \frac{1}{N+M}\left(\sum y_i + Z(t+\theta) + (M-Z)\Big(\theta - \frac{te^{-t/\theta}}{1-e^{-t/\theta}}\Big)\right).$$

(c) Numerical iterations. $N=4$, $\sum y_i = 9.5$, $M=6$, $t=2.5$, $Z=2$. Start with $\theta^{(0)}=2$:

$$\begin{aligned}
\mathbb{E}[X \mid E=1] &= 2.5 + 2 = 4.5,\\
\mathbb{E}[X \mid E=0] &= 2 - \frac{2.5\, e^{-1.25}}{1-e^{-1.25}} \approx 1.040,\\
\theta^{(1)} &= \frac{9.5 + 2\cdot 4.5 + 4\cdot 1.040}{10} = 2.266.
\end{aligned}$$

Continuing,

$$\begin{aligned}
\theta^{(2)} &\approx 2.412,\\
\theta^{(3)} &\approx 2.489,\\
\theta^{(4)} &\approx 2.529,\\
\theta^{(5)} &\approx 2.550.
\end{aligned}$$

(The sequence continues to converge to approximately 2.60.)

(d) Why the E-step is richer. Simply replacing every censored lifetime by the inspection time t would ignore the information contained in the failure indicators. Units that failed by t have conditional expectations strictly less than t, while units that survived have conditional expectations strictly greater than t. The E-step uses the exact conditional means that incorporate both pieces of information; a crude substitution by t would produce a biased and inefficient update.

7 Problem 7: Parameter-dependent support (tricky)

Now replace the exponential lifetime model in Problem 6 by

$$X \sim \text{Uniform}(0, \theta], \qquad \theta > 0.$$

Suppose the same two-experiment design is used, and at least one unit in experiment II survives beyond inspection time t.

(a) A tempting "EM-like" rule is to replace a missing X_i by its conditional mean and then plug these means into the complete-data MLE $\max\{Y_{\max}, X_{\max}\}$. Show that this produces the iteration

$$\theta^{(t+1)} = \max\Big\{Y_{\max},\ t + \frac{\theta^{(t)}}{2}\Big\}.$$

What fixed point does this suggest?

(b) Derive the observed-data likelihood when $Z \geq 1$ units survive past t, and show that, away from the boundary imposed by $Y_{\max}$, its interior maximiser is

$$\bar{\theta} = \frac{N+M}{N+M-Z}\, t.$$

(c) Explain precisely why the plug-in iteration in part (a) is not a valid EM algorithm.

(d) Suppose EM is initialised at $\theta^{(0)} > \max\{Y_{\max}, t\}$. Show that the genuine Q-function satisfies

$$Q(\theta \mid \theta^{(0)}) = \begin{cases} -(N+M)\log\theta + C, & \theta \geq \theta^{(0)}, \\ -\infty, & 0 < \theta < \theta^{(0)}, \end{cases}$$

for a constant C independent of θ. What does one EM update do?

Required Theory

Uniform distribution and parameter-dependent support. The density of Uniform$(0, \theta]$ is $1/\theta$ on $(0, \theta]$ and zero elsewhere. Consequently the complete-data likelihood is zero whenever any observation exceeds θ. The observed-data likelihood inherits the same hard constraint: θ must be at least as large as every observed lifetime and at least as large as t if any unit survived past t.

Why naive plug-in fails. EM requires the *expectation of the complete-data log-likelihood*, not the log-likelihood evaluated at expected complete data. When the support depends on the parameter the two operations do not commute.

Solution

(a) Tempting plug-in iteration. For a unit that survived past t the conditional distribution of X given $X > t$ (and $\theta > t$) is Uniform$(t, \theta]$. Its mean is $(t + \theta)/2$. Replacing every missing X_i by this mean and then taking the complete-data MLE $\max\{Y_{\max}, X_{\max}\}$ produces exactly the claimed recursion

$$\theta^{(t+1)} = \max\Big\{Y_{\max},\ t + \tfrac{1}{2}\theta^{(t)}\Big\}.$$

A fixed point of the interior map would satisfy $\theta = t + \theta/2$, i.e. $\theta = 2t$. (Whether this point is attainable depends on $Y_{\max}$.)

(b) Observed-data likelihood and interior maximiser. The contribution of the N exact observations from experiment I is θ^{-N} provided $\theta \geq Y_{\max}$. Each of the Z survivors contributes the survival probability $(\theta - t)/\theta = 1 - t/\theta$. Each of the $M - Z$ failures contributes the cdf value t/θ. Hence, for $\theta > \max\{Y_{\max}, t\}$,

$$L(\theta) = \theta^{-N}\Big(1 - \frac{t}{\theta}\Big)^{Z}\Big(\frac{t}{\theta}\Big)^{M-Z} = \theta^{-(N+M-Z)}(\theta - t)^{Z} t^{M-Z}.$$

Taking the logarithm and differentiating yields the critical point

$$\bar{\theta} = \frac{N+M}{N+M-Z}\, t,$$

provided this value exceeds $\max\{Y_{\max}, t\}$; otherwise the maximum is attained on the boundary.

(c) Why the plug-in is not EM. The complete-data log-likelihood contains the indicator that every $X_i \leq \theta$. Taking the expectation of this indicator (or of the whole log-likelihood) produces a function of θ that is $-\infty$ for all θ smaller than the essential supremum of the conditional distribution of the missing data. The plug-in procedure never examines this indicator; it merely substitutes conditional means into a formula that is valid only when the support constraint is already satisfied. Consequently the iteration does not maximise (nor even evaluate) the true Q-function.

(d) Behaviour of the genuine Q-function. If $\theta^{(0)} > \max\{Y_{\max}, t\}$ then, conditionally on the observed data and on $\theta^{(0)}$, every missing lifetime is almost surely less than or equal to $\theta^{(0)}$. Therefore the expected complete-data log-likelihood is finite only for $\theta \geq \theta^{(0)}$ and equals

$$Q(\theta \mid \theta^{(0)}) = -(N+M)\log\theta + C \qquad (\theta \geq \theta^{(0)}),$$

where C collects all terms that do not depend on θ. The maximiser of this Q on $[\theta^{(0)}, \infty)$ is the left endpoint $\theta^{(0)}$ itself. Thus a single EM update leaves the parameter unchanged: $\theta^{(1)} = \theta^{(0)}$. In other words, EM started above the observed maximum is immediately stuck; the algorithm cannot decrease θ below the value that is already known to dominate all data.

8 Problem 8: The ascent property of EM

Let

$$p_t(z) = p(z \mid y, \theta^{(t)}), \qquad Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{p_t}[\log p(y, Z \mid \theta)].$$

Assume all quantities below are well defined.

(a) Show the identity

$$\ell(\theta) - \ell(\theta^{(t)}) = Q(\theta \mid \theta^{(t)}) - Q(\theta^{(t)} \mid \theta^{(t)}) + D_{\mathrm{KL}}(p_t \| p(\cdot \mid y, \theta)),$$

where $\ell(\theta) = \log p(y \mid \theta)$.

(b) Deduce the ascent property

$$Q(\theta^{(t+1)} \mid \theta^{(t)}) \geq Q(\theta^{(t)} \mid \theta^{(t)}) \implies \ell(\theta^{(t+1)}) \geq \ell(\theta^{(t)}).$$

(c) Does monotone increase of the likelihood imply convergence to the global maximum? Explain briefly.

Required Theory

Jensen / KL representation of the incomplete-data log-likelihood. The fundamental identity underlying EM is obtained by writing

$$\log p(y \mid \theta) = \log p(y, z \mid \theta) - \log p(z \mid y, \theta)$$

and taking the expectation with respect to an arbitrary conditional distribution $p_t(z)$. The difference of the two expectations is precisely a Kullback–Leibler divergence, which is non-negative.

Solution

(a) Derivation of the identity. Start from the elementary decomposition

$$\log p(y \mid \theta) = \log p(y, Z \mid \theta) - \log p(Z \mid y, \theta).$$

Take the expectation of both sides with respect to $p_t(z) = p(z \mid y, \theta^{(t)})$:

$$\begin{aligned}\ell(\theta) &= \mathbb{E}_{p_t}[\log p(y, Z \mid \theta)] - \mathbb{E}_{p_t}[\log p(Z \mid y, \theta)] \\ &= Q(\theta \mid \theta^{(t)}) - \mathbb{E}_{p_t}[\log p(Z \mid y, \theta)].\end{aligned}$$

The same identity written at $\theta = \theta^{(t)}$ reads

$$\ell(\theta^{(t)}) = Q(\theta^{(t)} \mid \theta^{(t)}) - \mathbb{E}_{p_t}[\log p(Z \mid y, \theta^{(t)})].$$

Subtracting the two displays yields

$$\begin{aligned}\ell(\theta) - \ell(\theta^{(t)}) = Q(\theta \mid \theta^{(t)}) - Q(\theta^{(t)} \mid \theta^{(t)}) \\ + \Big(\mathbb{E}_{p_t}[\log p(Z \mid y, \theta^{(t)})] - \mathbb{E}_{p_t}[\log p(Z \mid y, \theta)]\Big).\end{aligned}$$

The expression in parentheses is exactly the Kullback–Leibler divergence

$$D_{\mathrm{KL}}\big(p_t \| p(\cdot \mid y, \theta)\big) = \mathbb{E}_{p_t}\Big[\log \frac{p_t(Z)}{p(Z \mid y, \theta)}\Big] \geq 0.$$

This establishes the claimed identity.

(b) Ascent property. Because the KL divergence is always non-negative we have

$$\ell(\theta) - \ell(\theta^{(t)}) \geq Q(\theta \mid \theta^{(t)}) - Q(\theta^{(t)} \mid \theta^{(t)}).$$

Choosing $\theta = \theta^{(t+1)}$ (any point that improves Q) immediately gives

$$\ell(\theta^{(t+1)}) - \ell(\theta^{(t)}) \geq Q(\theta^{(t+1)} \mid \theta^{(t)}) - Q(\theta^{(t)} \mid \theta^{(t)}) \geq 0.$$

(c) Global versus local maximisers. Monotone increase of $\ell(\theta^{(t)})$ guarantees that the sequence of likelihood values converges (to some finite limit if the likelihood is bounded above). It does *not* guarantee that the limit is a global maximiser. The observed-data likelihood of a mixture model, for example, is typically multimodal; EM may converge to a local maximum, a saddle point, or even a singularity, depending on the starting value. Global optimality requires additional arguments (convexity, uniqueness of the critical point, exhaustive search over multiple starts, etc.).

9 Problem 9: A Bernoulli mixture model

For $n = 1, \ldots, N$, observe a binary vector

$$Y_n = (Y_{n1}, \ldots, Y_{nM}) \in \{0, 1\}^M.$$

Assume a K-component mixture. Conditional on latent class $Z_n = k$, the M coordinates are independent Bernoulli variables with

$$\mathbb{P}(Y_{nm} = 1 \mid Z_n = k) = p_{km},$$

and $\mathbb{P}(Z_n = k) = \pi_k$, where $\sum_{k=1}^K \pi_k = 1$.

(a) Write the component probability $f_k(y_n)$ and the complete-data log-likelihood using indicators $Z_{nk} = \mathbf{1}\{Z_n = k\}$.

(b) Derive the posterior class responsibilities $\gamma_{nk}^{(t)}$.

(c) Derive the M-step updates for π_k and p_{km}.

(d) Explain in one or two sentences why the Bernoulli-mixture M-step has the same “expected counts divided by expected totals” structure as the Gaussian-mixture M-step.

Required Theory

Bernoulli product likelihood. Under conditional independence the component density is the product of Bernoulli pmfs:

$$f_k(y) = \prod_{m=1}^{M} p_{km}^{y_m}(1-p_{km})^{1-y_m}.$$

Complete-data log-likelihood for a discrete mixture. With class indicators Z_{nk},

$$\ell_c = \sum_n \sum_k Z_{nk} \log \pi_k + \sum_n \sum_k Z_{nk} \sum_m \big[y_{nm} \log p_{km} + (1-y_{nm}) \log(1-p_{km})\big].$$

The M-step consists of ordinary multinomial and Bernoulli MLEs computed with the expected indicators γ_{nk}.

Solution

(a) Component probability and complete-data log-likelihood.

$$f_k(y_n) = \prod_{m=1}^{M} p_{km}^{y_{nm}}(1-p_{km})^{1-y_{nm}}.$$

Using the indicators Z_{nk},

$$\begin{aligned}\ell_c(\theta; y, z) = &\sum_{n=1}^{N}\sum_{k=1}^{K} Z_{nk}\log \pi_k \\ &+ \sum_{n=1}^{N}\sum_{k=1}^{K} Z_{nk} \sum_{m=1}^{M}\big[y_{nm}\log p_{km} + (1-y_{nm})\log(1-p_{km})\big].\end{aligned}$$

(b) Posterior responsibilities. By Bayes' theorem

$$\gamma_{nk}^{(t)} = \mathbb{P}(Z_n = k \mid y_n, \theta^{(t)}) = \frac{\pi_k^{(t)} f_k(y_n; \mathbf{p}_k^{(t)})}{\sum_{j=1}^{K} \pi_j^{(t)} f_j(y_n; \mathbf{p}_j^{(t)})}.$$

(c) M-step updates. The expected complete-data log-likelihood separates into a multinomial term for the π_k and $K \times M$ independent Bernoulli terms for the p_{km}. Maximising yields

$$\begin{aligned}\pi_k^{(t+1)} &= \frac{1}{N}\sum_{n=1}^{N}\gamma_{nk}^{(t)}, \\ p_{km}^{(t+1)} &= \frac{\sum_{n=1}^{N}\gamma_{nk}^{(t)} y_{nm}}{\sum_{n=1}^{N}\gamma_{nk}^{(t)}}.\end{aligned}$$

(d) Structural analogy. In both the Bernoulli mixture and the Gaussian mixture the complete-data model belongs to an exponential family whose sufficient statistics are (weighted) counts or sums. The E-step replaces the unobserved indicators by their posterior expectations; the M-step then computes the ordinary maximum-likelihood estimators for those expected sufficient statistics. Consequently both algorithms produce updates of the form "expected success counts divided by expected trial totals" (or the analogous weighted mean and variance formulae for the Gaussian case).